

Soubhik Ghosh

GenAI & Machine Learning Engineer | LLMs | Agentic AI | RAG

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Professional Summary

GenAI & Machine Learning Engineer with **5+ years** of combined experience across HDFC Bank, PayU (Wibmo), and FICO, building production AI systems across Generative AI, LLM engineering, Agentic AI, RAG, Document AI, Computer Vision, biometrics, and intelligent automation. Strong hands-on expertise in **LLM orchestration, AI solution architecture, hybrid retrieval, AI microservices, MLOps, and full-stack AI applications**, with experience leading engineers and taking AI systems from experimentation to production. Recognized with the **HDFC Bank Silver Star Award** for AI technical excellence and presenting AI initiatives to the CEO and Board.

Technical Skills

Languages	Python, Java, TypeScript, JavaScript (ES6+), C/C++
GenAI / LLM	Vertex AI, LiteLLM, LangChain, LangGraph, RAG, Agentic AI, Prompt Engineering
ML / CV / NLP	PyTorch, Transformers, DistilBERT ML, RoBERTa, DeBERTa, LayoutLM, Donut, YOLOv8, Pix2Pix GAN, ArcFace, FaceNet, InsightFace, Wav2Vec2, Whisper, ECAPA-TDNN
Full Stack	ReactJS, Angular, Micro Frontends, ThreeJS, WebGL, FastAPI, Spring Boot, Node.js, REST APIs, Microservices
Data / Infra	PostgreSQL, MySQL, Couchbase, Redis, FAISS, BM25, Vector DB, YugabyteDB; Docker, Kubernetes, CI/CD, GCP, AWS, Kafka; Keycloak, FIDO2, OAuth, OWASP LLM Top 10

Experience

Data Scientist (Deputy Manager)
HDFC Bank

Nov 2024 - Present
Bengaluru, India

Generative Artificial Intelligence & Large Language Model Engineering

AI Banking PlatformLangGraph Agentic Orchestration

Stack: Python, LangGraph, FastAPI, Gemini, LiteLLM, ReactJS Micro Frontends, Keycloak, MFA, RAG

- Designed and drove the bank's **first agentic AI platform**, architecting a LangGraph-based orchestration layer that composes banking workflows from modular skills and agents while integrating **session management, MFA-aware agents, authentication, and downstream banking entities**.
- Integrated **25+ tools and banking capabilities** behind a unified tool-calling layer, enabling agents to securely invoke reusable services and workflows rather than relying on hard-coded conversational flows.
- Designed RAG capabilities over **public website and enterprise knowledge sources**, enabling agents to retrieve grounded, current information and combine it with authenticated banking workflows and tool responses.
- Designed and load-tested the platform for approximately **3 lakh concurrent users** ahead of its initial production rollout; the initiative was subsequently showcased at **Global FinTech Fest 2026**.
- Established modularity, authentication, observability, and code-quality standards across platform repositories.

AI Fabric: TradeOpsLLM Document Intelligence Platform

Stack: Python, FastAPI, Gemini, Vertex AI, ChandraOCR, LiteLLM, LayoutLM, BERT, Docker, CI/CD

- Architected and led the bank's **first production Python AI platform**: document classification/extraction microservices serving **800+ users pan-India** at **97% accuracy**, reducing manual processing time by **90%**.
- Built a **pluggable LLM architecture** using Vertex AI and LiteLLM, integrating Gemini with LayoutLM/BERT extraction, confidence scoring, prompt management, and an 11-API feedback loop; led 4 engineers and established CI/CD with **98%+ test coverage**.

Cheque Processing: Clearing HouseHigh-Throughput Document AI

Stack: Python, FastAPI, Gemini 3.5 Flash Lite, Vertex AI, LiteLLM, LayoutLM, RapidOCR, OpenCV

- Engineered a high-throughput cheque-processing engine combining **Gemini, LayoutLM, and Donut**, achieving **99%+ accuracy** at **10+ cheques/sec**; built a validation harness over **700K+ ICR reject records** to benchmark and improve production quality.

VerifyX: Audit, Credit & Retail AIRAG-Powered Verification Framework

Stack: ReactJS, FastAPI, Gemini, RAG, CAG Agent, LiteLLM

- Built a reusable audit and verification framework for VKYC and ECCS cheque-data workflows, with a **RAG/CAG chat agent** spanning **27,000+ documents**, reused across retail and credit.
- Optimized chunking and token strategies to improve retrieval efficiency and **reduce LLM consumption**; presented the framework and enterprise GenAI capabilities to the CEO and Board.

Narad AI & VERAEnterprise Agentic & Voice AI Interfaces

Stack: ReactJS, Micro Frontends, Llama Omni, FastAPI, LiteLLM, ThreeJS

- Built the **Narad AI email-agent** on a reusable micro-frontend architecture that categorises and responds to emails and raises tickets with **98%+ test coverage**, and developed the VERA voice-to-voice interface with real-time ThreeJS visualizations.

Machine Learning, Computer Vision & Biometrics

- Pay-by-Face & LOOK: Facial Biometrics** Real-Time Recognition + Anti-Spoofing
Stack: ArcFace, FaceNet, InsightFace, FAISS, Milvus, Custom Anti-Spoofing CNN, PyTorch
- Benchmarked **7 face-embedding models** and shipped **sub-100ms** recognition at **99.2% accuracy** under production load; built a custom anti-spoofing CNN using liveness, depth maps, and texture analysis with **false acceptance rate below 0.1%**.
- Signature Verification Pipeline** Document Forensics
Stack: YOLOv8, Pix2Pix GAN, Embedding Matching, Bicubic Interpolation, PyTorch
- Designed signature verification using custom YOLOv8 extraction, Pix2Pix GAN denoising, embedding similarity, and image interpolation.
- Vaani & Pay-by-Voice: Voice Biometrics** Speaker Verification + Voice-to-Voice
Stack: Indic Parler-TTS, Wav2Vec2, Whisper, ECAPA-TDNN/x-vector, PyTorch
- Built ECAPA-TDNN speaker verification reaching **97% accuracy**, with Wav2Vec2/Whisper, Indic Parler-TTS, and anti-spoofing voice forensics for multimodal Pay-by-Voice.

- Ankan: Image Labelling Platform** Active-Learning Annotation Tool
Stack: ReactJS, FastAPI, YugabyteDB, Active Learning, Quality Gating
- Built an active-learning annotation platform with quality gates; **100+ users** labelled **10,000+ images** for bank-wide ML training data.

- Pareekshana Automation Suite & AI OS** QA Automation + Platform Foundation
Stack: ReactJS MFE, RAG-based CAG, FastAPI, LLM Orchestration
- Redesigned the enterprise QA suite with Micro Frontends, RAG-based CAG, and LLM orchestration; led up to **10 engineers** and built AI-powered merchant-verification and onboarding compliance POCs.

- Software Engineer / Associate Software Engineer** Aug 2022 - Nov 2024
PayU (Wibmo) Bengaluru, India
- Owned the **Angular frontend end-to-end** for the Fraud Detection Risk Management Portal, including architecture, UI development, state management, performance optimization, and production deployment.
 - Architected and scaled **Spring Boot microservices**, developing **20+ REST APIs** with Couchbase and Redis caching and reducing response times by **50%** during high-traffic workloads.
 - Built secure authentication and API infrastructure using **FIDO2, FIDO UAM, Keycloak**, secure API gateways, Docker, Kubernetes, and Nginx; received PayU Quarterly Ace and ThankU Awards.

- Associate Software Engineer / Software Engineering Intern** Jun 2021 - Aug 2022
Fair Isaac Corporation (FICO) Bengaluru, India
- Built Python and Computer Vision pipelines for bank statements and identity documents; optimized C++/Python facial-recognition code toward **FRVT/NIST certification** and contributed to rPPG liveness-detection datasets and model development.

Open-Source Projects

- soweak: AI Security Framework (Python & TypeScript)** github.com/so-weak/soweak | PyPI & npm
Stack: Python, TypeScript, Transformers, RoBERTa, DeBERTa, NER, Sentence-Transformers, OWASP LLM Top 10
- Published OWASP-aligned middleware defending LLM boundaries across input, retrieval, tool calls, output, and streaming; shipped Python (PyPI) and TypeScript (npm) implementations with audit trails.
 - Trained RoBERTa/DeBERTa classifiers for prompt injection, jailbreak, PII/DLP, and toxicity detection, with LangChain, OpenAI, and Gemini adapters plus a red-team CLI.
- aakaar: NL to DAG Workflow Automation** github.com/so-weak/aakaar
Stack: Python, FastAPI, React, TypeScript, LLM Planner, WebSockets, SQLite, Chroma, PlaywrightMCP
- Built a multi-tenant platform that compiles natural-language tasks into typed DAG workflows through an LLM planner, with credential management, per-task audit, and remote execution over WebSockets.

Education

- B.Tech, Computer Science and Engineering**, KIIT, Bhubaneswar, **9.56 CGPA** 2018–2022

Awards & Certifications

- Silver Star Award** (HDFC Bank), AI technical excellence; **Quarterly Ace & ThankU Awards** (PayU); delegate, **Google I/O Connect Bengaluru 2025**.
- Google IT Support; From Data to Insights with Google Cloud; Human-Centered Design; **JLPT N4**.